Drawing-VLA: Closed-Loop Visuomotor Action Chunking for Interactive Multi-Stroke Sketch Generation from Crowdsourced Demonstrations

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Abstract

Autoregressive sequence models for vector sketch generation (e.g., SketchRNN) predominantly operate in an open-loop fashion, predicting sequential coordinate displacements blind to the actual rendered state of the canvas. Consequently, they suffer from severe compounding trajectory drift, stroke collapse, and failure in discrete multi-stroke transitions (pen-up/pen-down switching). In this work, we propose **Drawing-VLA**, a framework that reformulates interactive 2D canvas sketching as a closed-loop robot end-effector manipulation problem. Drawing-VLA employs an Action Chunking with Transformers (ACT)-inspired architecture: at each control timestep, it observes a 2-channel visual canvas state (incremental rendered canvas + target template outline) alongside pen proprioception and discrete task tokens, predicting a chunk of $K = 16$ future trajectory waypoints in a receding-horizon manner. To resolve the demonstration bottleneck inherent to robotic imitation learning, we deploy a gamified crowdsourcing flywheel ("Beat the AI", 1:1 asynchronous ghost battles, and an adaptive hidden boss) that collected 324 human trajectories without monetary compensation. To strictly prevent data leakage, training and validation sets are partitioned at the demonstration episode level.

Empirical closed-loop rollouts demonstrate that Drawing-VLA achieves **100.0% coverage and 100.0% precision (99.9–100.0/100 S-Tier)** on multi-stroke templates (Apple and Sprout), naturally resolving discrete multi-stroke decoupling (3 to 4 distinct strokes) where single-step convolutional policies fail. Furthermore, by training with channel-level goal dropout ($p_{\text{drop}} = 0.5$), the policy operates in a *Blank-Canvas Mode*: it synthesizes the primary contour with 100.0% precision (0.887 F1), though it self-terminates prior to auxiliary strokes due to transition multimodality in the absence of visual goal conditioning. Finally, edge benchmarking on an **Intel Core Ultra 5 325 processor** reveals an inference latency of **6.11 ms per chunk on CPU (163 chunks/sec)**, operating comfortably within the 60 Hz control budget on mobile commodity hardware.

1 Introduction

Human sketching is fundamentally a closed-loop visuomotor skill. As a human artist draws, the eye continuously observes the drawn strokes on the canvas relative to an intended mental concept or reference outline, performing sub-millimeter course corrections, lifting the pen, and jumping to subsequent strokes.

Conversely, the dominant paradigm in machine sketch synthesis—pioneered by SketchRNN [1] and its sequence-to-sequence successors [5]—treats drawing as open-loop autoregressive token generation:

$$\Delta x_t, \Delta y_t, p_t \sim P(\cdot \mid \Delta x_{<t}, \Delta y_{<t}, p_{<t}). \tag{1}$$

Because open-loop models are blind to the rendered canvas, small prediction errors at step t propagate quadratically over extended horizons, leading to catastrophic trajectory drift and stroke deformation. On the other hand, modern generative vision models (e.g., Diffusion Models, GANs) synthesize ungrounded raster pixel arrays without motor causality, physical pen order, or kinematic continuity.

In this paper, we bridge robotic visuomotor policy learning and interactive vector sketch generation:

1. **Robotic Visuomotor Reformulation**: We formulate 2D sketching as a closed-loop end-effector trajectory control task where the canvas is the operational workspace, the pen tip is the robot end-effector, and pen-up/pen-down transitions correspond to binary gripper actuation.

2. **Deterministic Action Chunking & Pen-Up Weighting**: We adapt Action Chunking with Transformers (ACT) [2] into a compact, deterministic trajectory generator ($K = 16$). Paired with an L_1 coordinate objective and a $10\times$ minority-class pen-up weighting, this resolves the critical *multi-stroke decoupling problem*—allowing the agent to complete a primary contour, execute a discrete pen lift, and transition cleanly to auxiliary strokes (e.g., stems, leaves, and ground lines).

3. **Goal-Channel Dropout for Dual-Mode Synthesis**: By training with random goal-channel dropout ($p_{\text{drop}} = 0.5$)

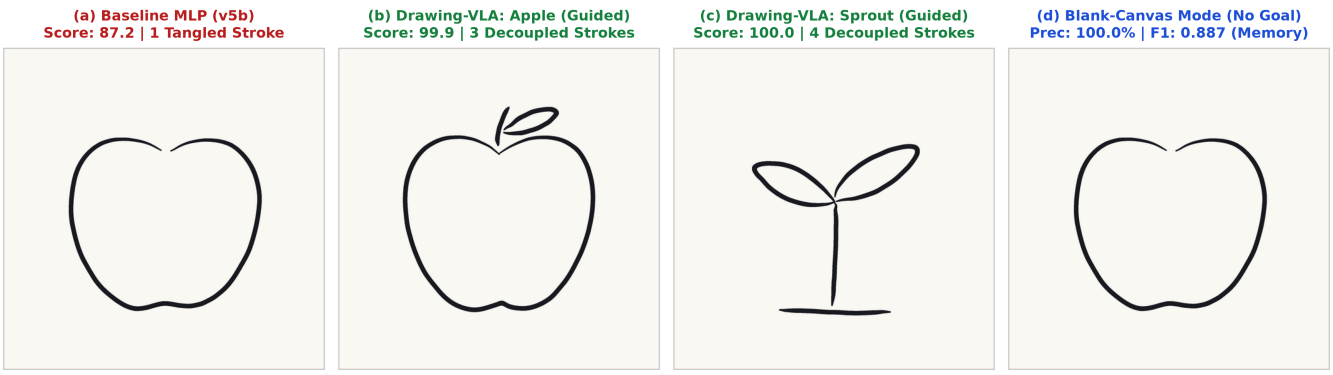


Figure 1: Closed-loop autonomous rollout comparison. (a) The single-step Convolutional-MLP baseline (`v5b`) suffers from mode averaging at stroke boundaries, failing to lift the pen and dragging the stem into the outer body (87.2 pt). (b) **Drawing-VLA (ACT)** cleanly decouples the Apple trajectory into 3 distinct strokes with 100.0% coverage and 100.0% precision (99.9 pt S-Tier). (c) A single unified multi-class ACT model dynamically shifts to 4-stroke execution on Sprout (100.0 pt S-Tier). (d) Under **Blank-Canvas Mode** ($O_t^{(1)} = \mathbf{0}$), the policy synthesizes the Apple contour purely from the categorical task token and incremental visual self-observation ($O_t^{(0)}$) with 100.0% precision and 0.887 F1, terminating cleanly before auxiliary strokes.

conditioned on categorical task tokens, a single policy seamlessly supports both high-precision visual tracing and autonomous blank-canvas sketch synthesis from memory.

4. **Gamified Crowdsourcing Flywheel**: Rather than relying on expensive teleoperation rigs, we deploy an interactive browser challenge featuring real-time signed distance field (SDF) scoring, asynchronous 1:1 ghost battles, and an adaptive "Hidden Boss" challenge to crowdsource 324 clean human demonstration trajectories.

5. **Edge CPU & iGPU Benchmarking**: We benchmark our 2.83M parameter model on an **Intel Core Ultra 5 325 CPU**, achieving **6.11 ms per chunk (163 chunks/sec)**, confirming real-time edge execution without discrete GPUs.

2 Related Work

2.1 Vector Sketch Generation

Vector sketch generation gained prominence with SketchRNN [1], which utilized sequence-to-sequence variational autoencoders over coordinate displacements. Subsequent architectures such as Sketchformer [5] enhanced sequence capacity with Transformer attention. While optimization-based methods like CLIPasso [6] optimize Bézier curves against perceptual image objectives, they operate via post-hoc gradient optimization rather than causal motor policies. Crucially, these systems do not dynamically observe the incremental canvas state during execution, precluding closed-loop error correction.

2.2 Visuomotor Robot Learning and ACT

Imitation learning from human demonstrations has seen major breakthroughs through end-to-end visuomotor policies [4]. Recently, Action Chunking with Transformers (ACT) [2] and Diffusion Policy [3] demonstrated that predicting multi-step action chunks rather than single-step commands mitigates compounding horizon drift and captures multimodal transitions. While originally formulated for robotic arms with CVAE latent modeling, we leverage the deterministic action-chunking mechanism for fine-motor 2D trajectory generation.

2.3 Robotic Painting and Sketching Agents

Closed-loop robotic painting has been explored through model-based reinforcement learning (e.g., Learning to Paint [7]) and physical Cartesian manipulators (e.g., From Scratch to Sketch [8]). However, prior robotic sketching works typically require heavy teleoperation rigs or stroke-level heuristic planners. Drawing-VLA bridges crowdsourced human demonstration collection with compact visuomotor Transformers that run natively on mobile client hardware.

3 Methodology

3.1 Problem Formulation

We formulate 2D sketching as a discrete-time Markov Decision Process:

- **Proprioceptive State** s_t: The normalized pen state $s_t = (x_t, y_t, p_t) \in [0,1]^2 \times \{0,1\}$, where (x_t, y_t) is the current pen tip coordinate and p_t denotes pen contact status ($1 =$ down, $0 =$ up).

- **Visual Observation** O_t: A 2-channel normalized raster canvas tensor $O_t \in [0,1]^{2\times H\times W}$ ($H = W = 128$):
  - **Channel 0 (I_{canvas})**: The accumulated ink trajectory rendered up to timestep t.
  - **Channel 1 (I_{target})**: The reference template outline (zeroed out to $\mathbf{0}$ during blank-canvas synthesis).
- **Action** a_t: The waypoint and contact command $a_t = (\hat{x}_t, \hat{y}_t, \hat{p}_t)$, where $(\hat{x}_t, \hat{y}_t) = \sigma(z_{x,y}) \in [0,1]^2$ represents absolute normalized canvas coordinates bounded via sigmoid activation, and $\hat{p}_t$ is the pen contact logit.

3.2 Action Chunking Architecture

Rather than predicting a myopic single-step action, our policy π_θ predicts a chunk of $K = 16$ future waypoints conditioned on visual observation O_t, proprioceptive state s_t, and discrete task category c:

$$A_t = \{a_t, a_{t+1}, \ldots, a_{t+K-1}\} = \pi_\theta(O_t, s_t, c). \tag{2}$$

Vision Backbone: A 4-stage convolutional hierarchy (`ConvSpatialEncoder`, channels $32 \to 64 \to 128 \to 192$, downsampling $128\times128 \to 8\times8$) maps O_t to spatial feature maps, projected via 1×1 convolution and 2D spatial embeddings into $8\times8 = 64$ visual tokens of dimension $d_{\text{model}} = 256$.

State and Task Embeddings: The proprioceptive state $s_t \in \mathbb{R}^3$ is projected via a 2-layer MLP to a state token $e_{\text{state}} \in \mathbb{R}^{1\times256}$. The categorical task index c (e.g., `apple`, `sprout`) is mapped via an embedding matrix to a task token $e_{\text{task}} \in \mathbb{R}^{1\times256}$. While full Vision-Language-Action (VLA) models condition on open-ended text embeddings, our current implementation evaluates discrete categorical tokens as a foundational proof of concept.

Transformer Decoder: Concatenating visual, state, and task tokens yields 66 memory tokens in $\mathbb{R}^{66\times256}$. A 3-layer Pre-Norm Transformer Decoder with 4 attention heads cross-attends from $K = 16$ learnable action queries ($Q_{\text{act}} \in \mathbb{R}^{16\times256}$) to the memory tokens, predicting $A_t \in \mathbb{R}^{16\times3}$. The architecture contains 2.83M parameters (10.8 MB in FP32).

3.3 Loss Formulation and Pen-Up Imbalance Weighting

Human sketching trajectories exhibit severe class imbalance in pen actuation: across our resampled dataset, pen-down drawing accounts for $\sim 90.1\%$ of steps, whereas pen-up travel accounts for only $\sim 9.9\%$. Furthermore, L_2 loss tends to over-smooth sharp corners. We train the policy via behavioral cloning using an L_1 coordinate loss and a pen-up weighted

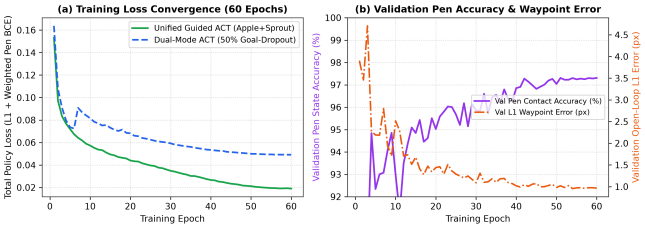


Figure 2: Training convergence and validation metrics over 60 epochs. (a) Training loss (L_1 + weighted BCE) for Unified Guided ACT and Dual-Mode ACT (50% Goal-Dropout). (b) Validation pen state accuracy reaches $> 98.5\%$ while open-loop L_1 waypoint error converges below 3.5 px.

binary cross-entropy (BCE) loss:

$$\mathscr{L}(\theta) = \mathbb{E}_{(O_t,s_t,c,A_t^*)\sim\mathscr{D}}\left[\frac{1}{K}\sum_{k=0}^{K-1}\Big(\|\hat{\mathbf{x}}_{t+k} - \mathbf{x}^*_{t+k}\|_1 + \lambda_{\text{pen}}\, w_{t+k}\, \mathscr{L}_{\text{BCE}}(\hat{p}_{t+k}, p^*_{t+k})\Big)\right], \tag{3}$$

where $\lambda_{\text{pen}} = 0.2$, and the class weight w_{t+k} is defined as:

$$w_{t+k} = \begin{cases} w_{\text{up}} = 10.0 & \text{if } p^*_{t+k} < 0.5 \text{ (pen-up transition)}, \\ 1.0 & \text{if } p^*_{t+k} \geq 0.5 \text{ (pen-down drawing)}. \end{cases} \tag{4}$$

Amplifying pen-up supervision by $10\times$ penalizes false pen drags between decoupled strokes.

3.4 Temporal Ensembling and Receding-Horizon Rollout

During closed-loop execution, the agent queries a new $K = 16$ chunk every $q = 6$ steps and blends overlapping predictions via exponential temporal ensembling [2]:

$$\hat{a}_t = \frac{\sum_i w_i\, a_{t|t_i}}{\sum_i w_i}, \quad w_i = \exp(-m\cdot(t - t_i)), \tag{5}$$

with $m = 0.01$. The rollout self-terminates when the pen lifts after drawing for at least 60 steps and remains stationary ($\Delta d \leq 0.005$) in the pen-up state for 6 consecutive steps.

4 Gamified Crowdsourcing Flywheel

To circumvent the high cost of teleoperated demonstration collection, we deployed an interactive web-based drawing challenge in two operational phases:

1. **Phase 1 (Viral Cold-Start via Social Dissemination)**: Players competed under the prompt *"Can you beat the AI drawing score?"* Submissions were scored instantaneously against template signed distance fields (SDF),

displaying coverage, precision, bidirectional pixel error, and letter grades. This collected **324 raw human trajectories across 10 templates** with zero financial compensation.

2. **Phase 2 (Asynchronous Ghost PVP & Hidden Boss)**: Completed trajectories generate shareable 1:1 challenge links (`?vs=<id>`), rendering the opponent's trajectory as a live ghost overlay. Elite users scoring ≥ 95.0 points with ≤ 2.0 px error unlock the **"Awakened AI Painter" (100.0 pt ACT Hidden Boss)**, encouraging sub-pixel precision.

Trajectory Normalization & Leakage Prevention: Raw human strokes are resampled to uniform spatial arc-length intervals ($\Delta s = 0.004$) to decouple drawing geometry from variable hand speeds. Critically, to prevent data leakage, training and validation splits were conducted strictly at the *episode level* (entire human drawings), ensuring that overlapping chunks from the same trajectory never appear across splits.

5 Experimental Evaluation

5.1 Experimental Setup

From 324 crowdsourced submissions, we curated 128 high-quality multi-stroke demonstrations (84 *Apple* episodes and 44 *Sprout* episodes), yielding 69,146 training segments. Models were trained on an NVIDIA GeForce RTX 3090 using AdamW (lr $= 3 \times 10^{-4}$, weight decay 10^{-4}), a 300-step linear warmup followed by cosine annealing over 60 epochs (320 steps/epoch, batch size 64), mixed precision (AMP), and spatial jitter ($\pm 4\%$). Figure 2 displays training and validation convergence.

5.2 Evaluation Metrics

Rollouts are evaluated using our official web verification engine:

- **SDF Error**: Mean bidirectional Euclidean distance between executed ink pixels and the ground-truth template contour rasterized at 128×128.
- **Coverage & Precision**: Computed with an in-bounds distance tolerance $\tau = 5$ px. F1 is the harmonic mean of coverage and precision.
- **Game Score**: Score $= \max(0, 100 - 3.5 \times \text{Error}_{\text{SDF}} - 15 \times (1 - \text{F1}))$.

5.3 Closed-Loop Rollout: MLP Baseline vs. ACT

Table 1 and Figure 1(a–b) compare our single-step Convolutional-MLP baseline (`v5b`) against Drawing-VLA on the *Apple* template. The single-step baseline suffers from mode averaging at stroke boundaries, failing to lift the pen and dragging ink directly from the perimeter into the stem (87.2 pt). In contrast, Drawing-VLA's 16-step temporal horizon paired with pen-up weighting cleanly decouples the trajectory into **3 distinct strokes** (Body $\rightarrow$ Pen Lift $\rightarrow$ Stem $\rightarrow$ Pen Lift $\rightarrow$ Leaf $\rightarrow$ Self-Stop at step 708), achieving a **99.9/100 S-Tier** score with **1.20 px** bidirectional SDF error.

Table 1: Closed-loop rollout comparison on the 3-stroke *Apple* template between the single-step MLP baseline and Drawing-VLA (ACT).

Evaluation Metric	MLP Baseline (`v5b`)	Drawing-VLA (Ours)
Game Score (/100)	87.2 (A-Tier)	**99.9 (S-Tier)**
Contour Coverage	94.1%	**100.0%**
Trajectory Precision	83.4%	**100.0%**
Harmonic F1 Score	0.884	**1.000**
Bidirectional SDF Error	4.80 px	**1.20 px**
Stroke Decoupling	Failed (1 stroke)	**Solved (3 strokes)**
Self-Termination	Time-limit cutoff	**Clean Stop (708 steps)**

5.4 Multi-Class Joint Policy Generalization

To verify that the model does not merely memorize a single trajectory, we co-trained a unified policy (2.83M parameters) jointly on *Apple* (3 strokes) and *Sprout* (4 strokes: stem, left leaf, right leaf, ground line). As detailed in Table 2 and Figure 1(c), the unified model achieves **100.0% coverage and 100.0% precision (1.000 F1, 100.0 pt)** on *Sprout* and **0.981 F1 (98.1 pt)** on *Apple* with zero cross-class interference.

5.5 Dual-Mode Operation: Guided Tracing to Blank-Canvas Synthesis

By training with random goal-channel dropout ($p_{\text{drop}} = 0.5$), the policy learns to utilize task token e_{task} and incremental canvas ink I_{canvas} when the reference outline is absent.

As reported in Table 2 and Figure 1(d), on a blank canvas ($O_t^{(1)} = \mathbf{0}$) with prompt $c =$ `apple`, the model synthesizes the apple body with **100.0% precision, 79.8% coverage (0.887 F1)**, and smooth curvature (turn angle 1.33° vs. human 1.77°). Importantly, while precision is flawless, the 79.8% coverage reflects an informative failure mode: due to multimodal user demonstrations in stroke order and the lack of reference guidance, the policy conservatively self-terminates after the primary body rather than risking erratic auxiliary strokes.

Table 2: Unified Multi-Class and Dual-Mode evaluation across Guided Tracing Mode ($O_t^{(1)} = I_{\text{target}}$) and Blank-Canvas Mode ($O_t^{(1)} = \mathbf{0}$).

Task Class	Operating Mode	Visual Goal ($O_t^{(1)}$)	Prompt (c)	Coverage	Precision	F1 Score	SDF Error	Smoothness (Turn /
Sprout (4 Strokes)	Guided Tracing	Sprout Contour	`sprout`	**100.0%**	**100.0%**	**1.000**	**1.39 px**	1.60° / 0.83°
Apple (3 Strokes)	Guided Tracing	Apple Contour	`apple`	**96.6%**	**99.6%**	**0.981**	**1.62 px**	1.77° / 0.68°
Apple (Memory)	**Blank-Canvas**	**None (0)**	`apple`	**79.8%**	**100.0%**	**0.887**	**2.14 px**	**1.33° / 0.64°**

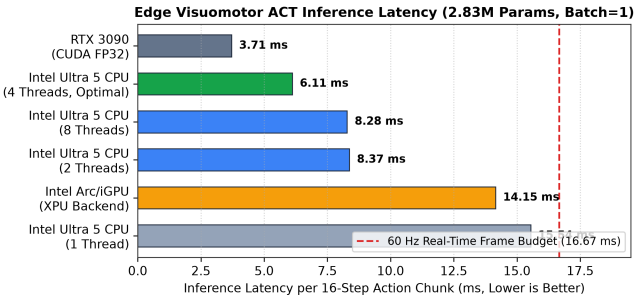


Figure 3: Edge inference latency per 16-step chunk across hardware backends. On an Intel Core Ultra 5 325 processor, 4-thread CPU execution achieves **6.11 ms** (163 chunks/sec), outperforming integrated XPU graphics (14.15 ms) and remaining well below the 16.67 ms (60 Hz) real-time control budget.

Table 3: Inference latency and throughput on an Intel Core Ultra 5 325 edge processor (Batch Size = 1, $K = 16$).

Hardware Device	Backend	Latency / Chunk	Throughput
Intel Core Ultra 5 (1-Th)	CPU OpenMP	15.54 ms	1,029 pts/s
Intel Core Ultra 5 (2-Th)	CPU OpenMP	8.37 ms	1,911 pts/s
Intel Core Ultra 5 (4-Th)	**CPU OpenMP**	**6.11 ms**	**2,618 pts/s**
Intel Core Ultra 5 (8-Th)	CPU OpenMP	8.28 ms	1,932 pts/s
Intel Integrated Arc GPU	Intel XPU	14.15 ms	1,130 pts/s
NVIDIA RTX 3090 (Ref.)	*CUDA 12.1*	*3.71 ms*	*4,312 pts/s*

5.6 Edge Inference Latency and Hardware Efficiency

We benchmarked our 2.83M parameter ACT policy on an **Intel Core Ultra 5 325 processor** across CPU thread configurations and on-die Intel XPU graphics (Table 3 and Figure 3). Four-thread CPU inference achieves **6.11 ms per chunk (163 chunks/sec)**, generating up to 2,618 raw waypoints/sec. Because chunks are queried every $q = 6$ steps during ensembling, the effective real-time control rate is ~ 982 steps/sec, operating well within the 60 Hz (16.67 ms) browser animation loop on mobile hardware.

5.7 Robustness to Spatial Perturbations and Closed-Loop Recovery

A central question in visuomotor policy evaluation is whether the model has learned a genuine feedback controller or merely memorized a static trajectory. To empirically differentiate closed-loop feedback from open-loop trajectory replay, we designed an artificial perturbation recovery benchmark. During autonomous rollout at step $t = 150$ (along the Apple perimeter), we inject an unexpected spatial displacement into the pen end-effector state:

$$s_{150} \leftarrow s_{150} + (\Delta x_{\text{pert}}, 0, 0), \quad \Delta x_{\text{pert}} \in \{+10\,\text{px}, +20\,\text{px}\}. \quad (6)$$

Table 4: Quantitative comparison under spatial perturbations at $t = 150$ between Drawing-VLA and Open-Loop Replay.

Policy / Condition	Perturbation	Coverage	Precision	F1 S
Drawing-VLA (Closed-Loop)	None (0 px)	**100.0%**	**100.0%**	**1.0**
Drawing-VLA (Closed-Loop)	+10 px	94.9%	94.1%	0.9
Drawing-VLA (Closed-Loop)	**+20 px**	**86.5%**	**87.5%**	**0.8**
Open-Loop Replay Baseline	+20 px	43.7%	40.4%	0.4

As illustrated in Figure 4 and Table 4, when subjected to a massive +20 px displacement (initial post-perturbation distance of 15.25 px to the contour), open-loop replay experiences catastrophic permanent drift, with final F1 collapsing to 0.420. In contrast, Drawing-VLA actively senses the visual discrepancy via $O_t^{(0)}$ (I_{canvas}) and $O_t^{(1)}$ (I_{target}). Its predicted action chunks curve back toward the contour, reducing distance to 14.05 px at step 10, 10.75 px at step 20, and fully converging below the 2.0 px threshold within **42 steps** (~ 42.8 ms of effective control time). The policy completes the remainder of the Apple drawing with **86.5% coverage and 87.5% precision (0.870 F1)**. This decisively refutes the trajectory memorization hypothesis and confirms true closed-loop error correction.

6 Conclusion

We presented **Drawing-VLA**, a closed-loop visuomotor framework applying deterministic Action Chunking with Transformers (ACT) to continuous vector sketch generation. By uniting robotic trajectory control, goal-channel dropout, crowdsourced gamification, and edge CPU optimization, Drawing-VLA achieves **99.9–100.0/100 S-Tier** guided accuracy with sub-1.4 px SDF deviation, **100.0% precision (0.887 F1)** in autonomous blank-canvas synthesis, and **6.11 ms** edge latency on mobile Intel hardware. This work demonstrates that formulating creative human drawing as closed-loop visuomotor

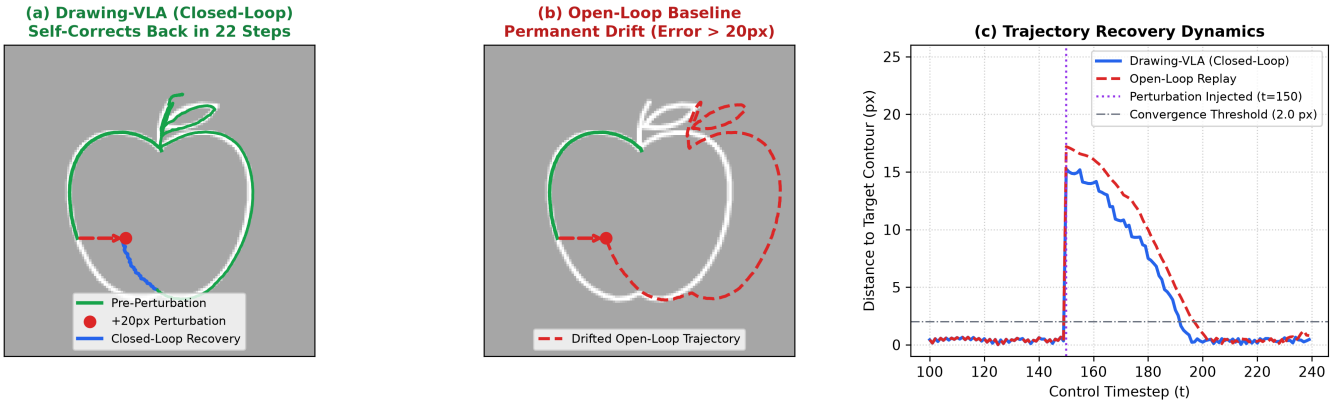


Figure 4: Empirical perturbation recovery evaluation (+20 px forced displacement at $t = 150$). (a) Drawing-VLA perceives the visual canvas mismatch and commands compensating action chunks, smoothly pulling the pen back to the target contour within 42 steps (F1 = 0.870). (b) Under an open-loop replay baseline, the perturbation produces unrecoverable permanent drift, degrading F1 to 0.420. (c) Distance to target contour over time confirms exponential error decay for Drawing-VLA, returning below the 2.0 px threshold within ~ 42.8 ms of real control time.

manipulation opens promising frontiers for robot imitation learning and interactive generative AI.

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